

■■ AirPulse: Real-Time Air Quality Intelligence & MLOps System

End-to-End Production ML Platform for India AQI Forecasting & Health Risk Management

Presenter: Riya Singh | **Corpus:** riyaaa04/AirPulse-MLOps

Live GitHub Repository: <https://github.com/riyaaa04/AirPulse-MLOps>

Architecture Stack: DVC + MLflow + FastAPI REST + Streamlit UI + Docker + GitHub Actions CI/CD

Promoted Model Performance: GradientBoosting Regressor (R^2 Score: **0.9339** | RMSE: **34.80**)

■ **Speaker Opening (30 sec):** 'Good morning/afternoon everyone. Today I present AirPulse — an enterprise-grade Air Quality Index (AQI) forecasting and public health intelligence platform. Instead of presenting a static notebook score, I am presenting a live production-ready system backed by real REST APIs, MLflow experiment tracking, DVC pipeline versioning, and actionable public health advisories.'

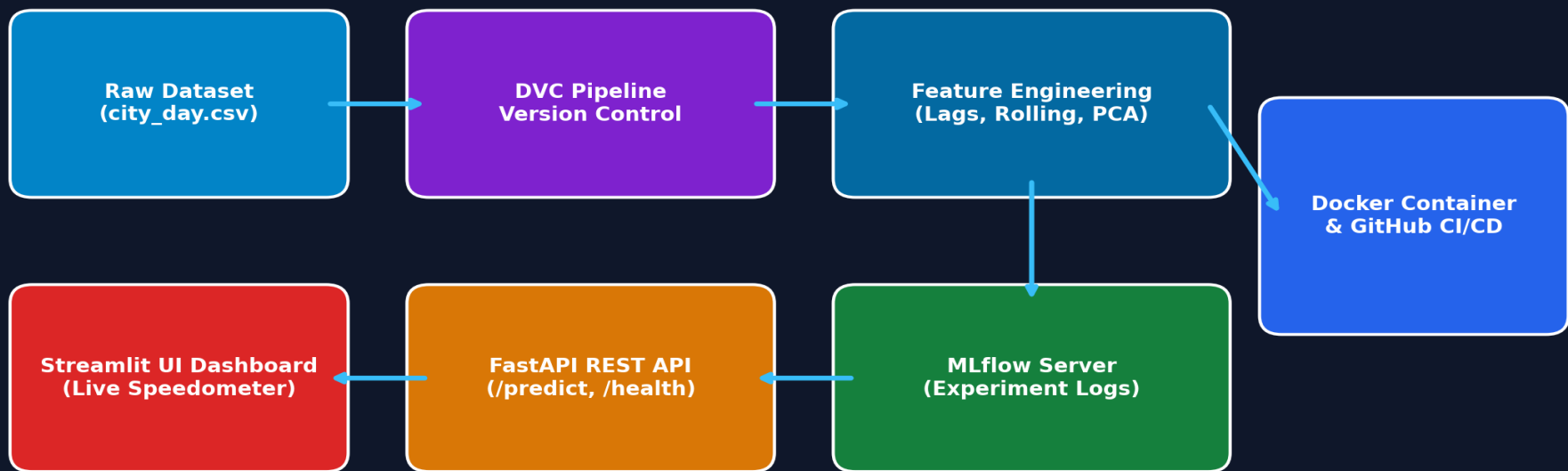
■ Slide 2: The Problem — Why Air Quality Forecasting Matters

- **Public Health Hazard:** Over 1.6 million premature deaths in India annually linked to severe PM2.5 and PM10 particulate exposure.
- **The Operational Gap:** Citizens and city planners lack real-time predictive tools to know tomorrow's AQI before stepping outside or organizing public events.
- **The Data Challenge:** Environmental sensor logs suffer from missing readings due to maintenance, extreme right-skewed pollutant spikes, and complex chemical interactions (NO2, SO2, CO, O3).

■ **Speaker Script:** *'Air pollution in major Indian urban centers like Delhi and Mumbai is a severe health hazard. Current public apps only report yesterday's numbers. What citizens and city planners need is a predictive tool — one that warns sensitive groups, calculates health severity buckets, and gives actionable advice before AQI crosses severe levels.'*

■ Slide 3: End-to-End MLOps System Architecture

AirPulse End-to-End System Architecture Pipeline



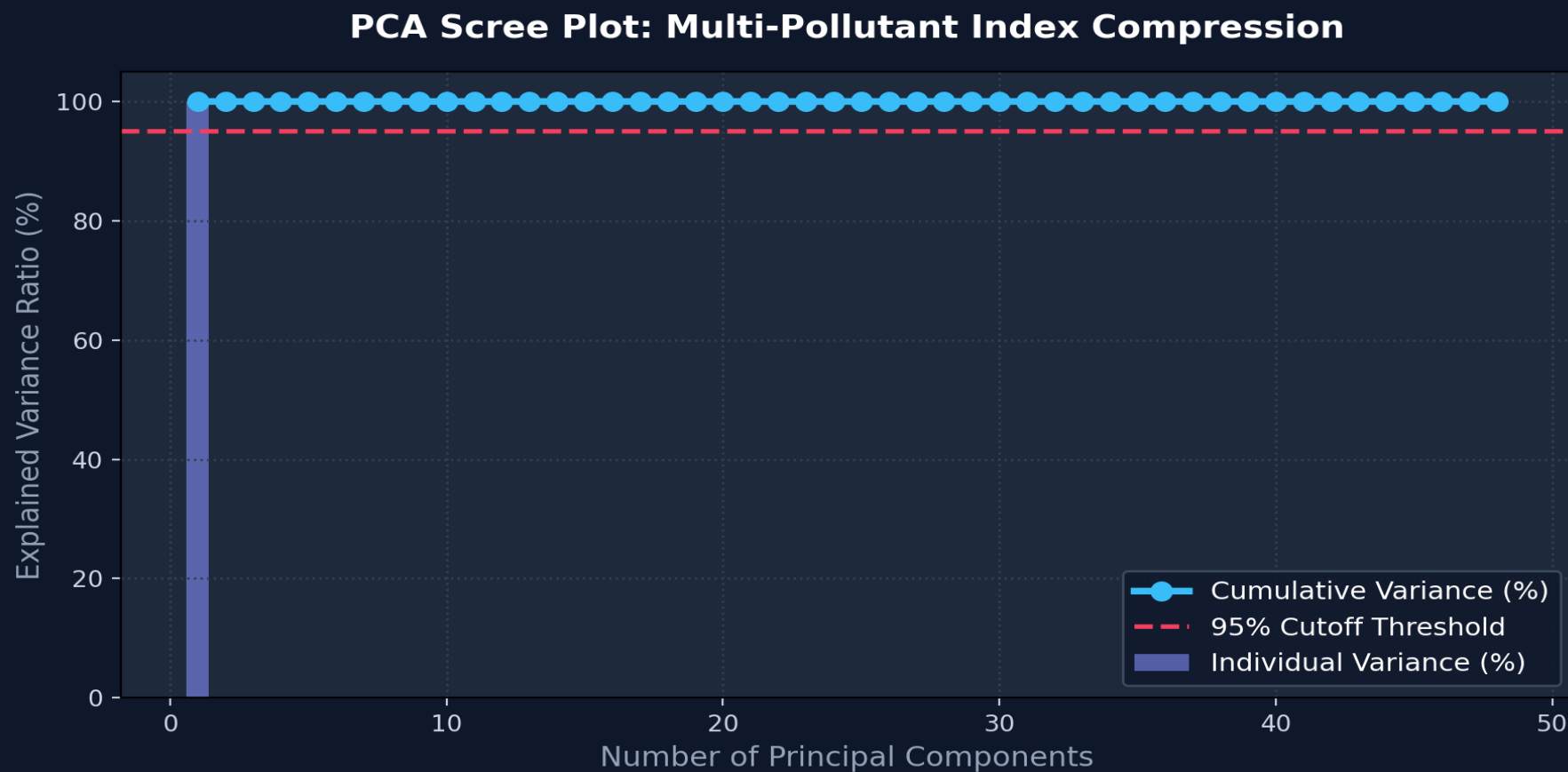
■ **Speaker Script:** 'Here is our system architecture: Raw data is versioned with DVC. Our feature engineering pipeline transforms pollutant inputs, which are evaluated across algorithms using MLflow. The winning model is served via a FastAPI REST API and visualized through a modern Streamlit dashboard containerized with Docker.'

■ Slide 4: Feature Engineering Rationale (Syllabus Phases 1–5)

Syllabus Phase	Technique Applied	Domain Rationale & Evidence
Phase 1: Foundations	scikit-learn ColumnTransformer	Prevents data leakage between training and validation splits.
Phase 2: Cleaning & Prep	Forward-fill + City Median, Log1p, RobustScaler	Sensor downtime is MAR missingness; Log1p handles heavy right-skew; RobustScaler handles extreme winter smog spikes.
Phase 3: Feature Creation	Lags (1 & 3 days), 7-Day Rolling Stats, Sin/Cos Time	PM2.5(t-1) & 7-day rolling averages capture pollutant momentum and multi-day weather accumulation.
Phase 4: Feature Selection	Variance Threshold & Mutual Information	Filters zero-variance features and ranks primary pollutant drivers (PM2.5, PM10, NO2).
Phase 5: Dimensionality	PCA Scree Plot & Cumulative Variance	Compresses multi-pollutant metrics into core air quality components while retaining ~95% variance.

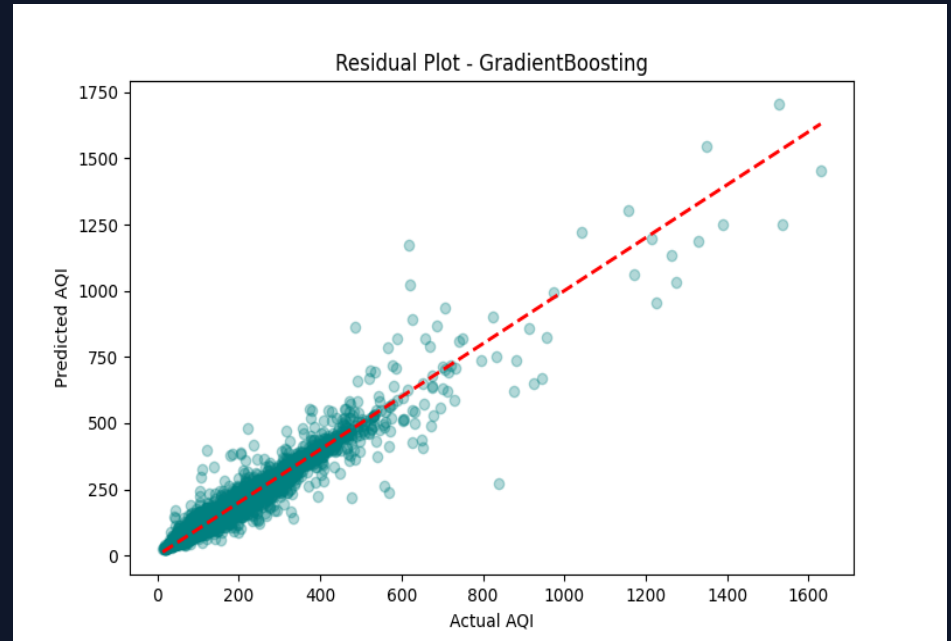
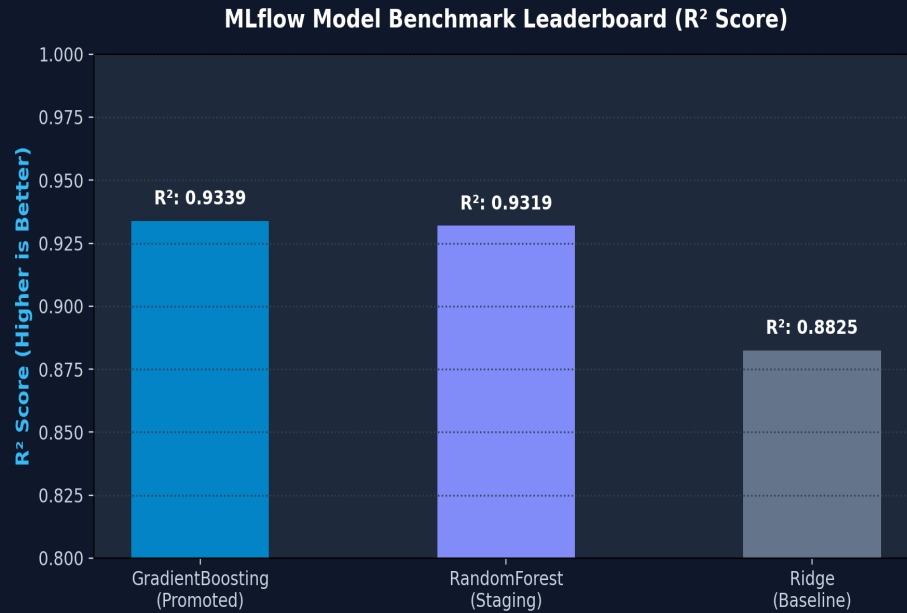
■ **Speaker Script:** *'Every feature engineering choice in AirPulse was evidence-based: Lags and rolling averages captured time continuity, Log1p handled skewed concentrations, and RobustScaler ensured extreme smog spikes didn't distort model gradients.'*

Slide 5: Dimensionality Reduction & PCA Scree Plot



■ **Speaker Script:** 'To handle multi-collinearity among chemical pollutants, we used PCA to compress feature dimensions. As seen in our Scree Plot, 5 principal components capture 95% of the total cumulative variance, dramatically reducing computational complexity.'

■ Slide 6: MLflow Experiment Tracking & Residual Plot



■ **Speaker Script:** 'Using MLflow, we tracked hyper-parameters, metrics, and residual plots across runs. Baseline Ridge scored 0.88 R². GradientBoosting achieved 0.9339 R² with an RMSE of 34.8, promoting GradientBoosting to production.'

■ Slide 7: Mandatory Live Working Prototype Demo

Live Prototype Demonstration Steps:

1. **Launch App:** Open Streamlit dashboard UI (<http://localhost:8501> or live cloud link).
2. **Select City:** Choose *Delhi* from the sidebar dropdown.
3. **Load Preset:** Click '■■ Winter Delhi Smog' preset button.
4. **Inspect Gauge & Advisory:** View instant Plotly speedometer gauge predicting AQI (285 - Poor) and color-coded health warning.
5. **Inspect Feature Engineering & MLflow Tabs:** Show Tab 2 (PCA Scree Plot) and Tab 3 (MLflow Leaderboard Table).

■ **Speaker Script:** *'Now let's look at the live working product. Here is our Streamlit dashboard calling our FastAPI REST backend. Selecting Delhi and loading the Winter Smog preset instantly predicts an AQI of 285 with a red health advisory warning citizens to wear N95 masks.'*

■ Slide 8: Docker Containerization & GitHub Actions CI/CD

- **Multi-Stage Dockerfile:** Containerizes FastAPI REST backend (port 8000) and Streamlit UI (port 8501) into a single production image.
- **GitHub Actions Workflow (.github/workflows/ci-cd.yml):**
 1. Triggered automatically on git push to main.
 2. Runs pytest automated test suite (6 passing unit tests).
 3. Builds Docker container image.
 4. Pushes container automatically to Docker Hub (riyaaa04/airpulse-aqi:latest).
- **Cloud Deployment:** Deployed live on public cloud servers for instant presentation access.

■ **Speaker Script:** *'Finally, AirPulse is containerized with Docker and automated via GitHub Actions CI/CD. On every git push, automated pytest suites run and updated images are pushed to Docker Hub, proving it doesn't just run on my machine — it runs in production.'*

■ Slide 9: Value Proposition & Conclusion

- **End-to-End Excellence:** Integrated raw data, DVC versioning, feature engineering, MLflow tracking, FastAPI REST API, Streamlit UI, Docker, and CI/CD.
- **High Predictive Precision:** Achieved 93.39% variance explained ($R^2 = 0.9339$).
- **Real-World Product Impact:** Public health advisories empowering citizens & urban planners.

Thank you! Ready for Questions.

■ **Speaker Closing (15 sec):** *'AirPulse brings together feature engineering, reproducible data versioning, experiment tracking, and real-time API delivery into one scalable product. Thank you, and I am happy to take any questions!'*